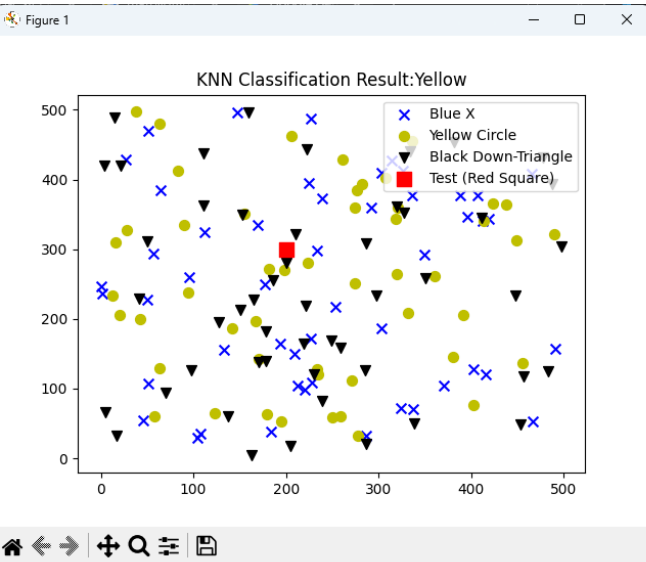




```
1.py X
14 plt.scatter(blue_x[:,0], blue_x[:,1], s=50, c='b', marker='x', label='Blue X')
15
16 yellow_o = trains[labels.ravel() == 1]
17 plt.scatter(yellow_o[:,0], yellow_o[:,1], s=50, c='y', marker='o', label='Yellow Circle')
18
19 black_v = trains[labels.ravel() == 2]
20 plt.scatter(black_v[:,0], black_v[:,1], s=50, c='k', marker='v', label='Black Down-Triangle')
21
22 test = np.array([[200, 300]]).astype(np.float32)
23 plt.scatter(test[:,0], test[:,1], s=100, c='r', marker='s', label='Test (Red Square)')
24
25 k = 7
26 knn = cv2.ml.KNearest_create()
27 knn.train(trains, cv2.ml.ROW_SAMPLE, labels)
28
29 ret, results, neighbours, dist = knn.findNearest(test, k=k)
30
31 class_names = {0.0: "Blue", 1.0: "Yellow", 2.0: "black"}
32 result_name = class_names[results[0][0]]
33
34 plt.title(f"KNN Classification Result:{result_name}")
35 plt.legend(loc='upper right')
36
37 print(f"鄰居標籤為: {neighbours.ravel()}")
38 print(f"最終判定為: {result_name}")
39
40 plt.show()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\User\Desktop\新增資料夾> python -u "c:\Users\User\Desktop\新增資料夾\1.py"
鄰居標籤為: [0. 2. 1. 0. 1. 0. 0.]
最終判定為: Blue
PS C:\Users\User\Desktop\新增資料夾> python -u "c:\Users\User\Desktop\新增資料夾\1.py"
鄰居標籤為: [2. 2. 1. 1. 1. 0. 2.]
最終判定為: Yellow
```

powerShell
Code